

The Collapse of Heterogeneity in Silicon Philosophers

Yuanming Shi, Adobe Inc.
Andreas Haupt, Stanford University

Question

Can an LLM stand in for a population of minds?

Setup

- 277 PhilPeople philosophers
- 100 PhilPapers 2020 questions
- 7 profile-conditioned LLMs
- Temperature 0; Accept $\rightarrow$ Reject coded onto $[0, 1]$
- Checked against PhilPapers 2020 ($N = 1,785$)

What we compare

- **Spread:** per-question variance
- **Shape:** correlations and PCA axes
- **Stereotypes:** specialist–position links

The test

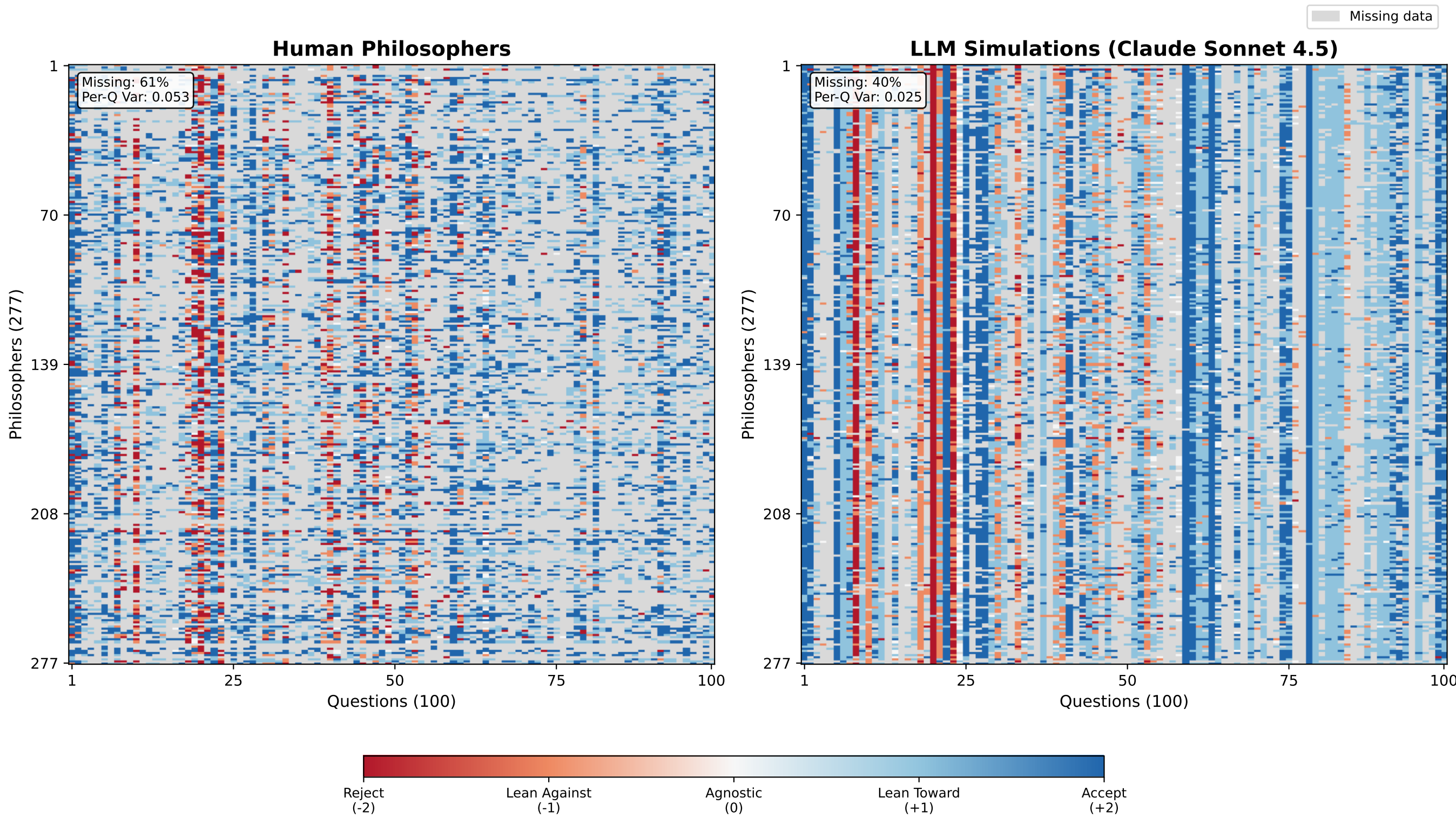
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A social world model— a generative model meant to stand in for a population of human minds— is adequate only if the distribution it induces over **agent-types** matches the population it represents.



Take a picture for the paper and data explorer.

Human vs. Claude



Each cell is one philosopher on one question. Humans: Per-Q Var = 0.053, 61% missing. Claude: 0.025, 40% missing.

The induced world is flatter

Across seven models, variance is $1.4\text{--}2.4\times$ lower than humans. Claude is uniform on 10 questions.

And mis-shaped

Human disagreement spans several independent axes. Models cluster by surface topic. DPO improves correlations but does not restore the distribution.

Stereotypes fill the gap

Where humans show no specialist effect, models invent a large one:

- Biology $\rightarrow$ biological identity: +11 pp vs +43 pp
- Biology $\rightarrow$ psychological identity: +4 vs -66 pp
- Ancient $\rightarrow$ Aristotelian reason: +2 vs +69 pp